



3D MRI Brain Age Prediction using Gaussian Splatting

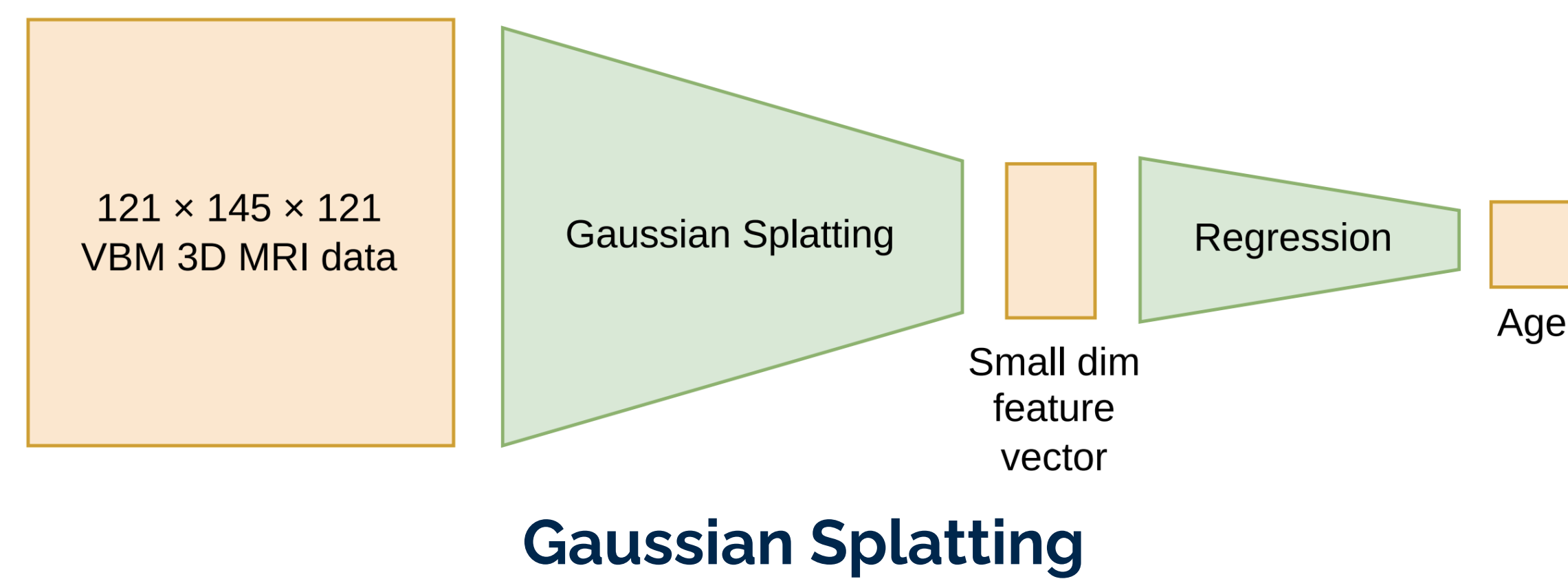
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EECS 545

Motivation

- Machine learning models can analyze structural brain MRI data to predict a subject's biological brain age.
- A predicted brain age significantly higher than chronological age indicates an atypical aging trajectory.
- This discrepancy, known as the brain age gap, is an early warning sign for cognitive decline and neurodegenerative diseases such as Alzheimer's, Parkinson's, and Huntington's disease [1, 3].
- We train an age regression model on the OpenBHB [2] dataset of MRI scans from healthy individuals to predict the brain age.

Pipeline



- Voxels are modeled as weighted sums of learnable 3D Gaussians [4].
- For a spatial location $\mathbf{j} = (x_j, y_j, z_j)$ the voxel intensity $I(\mathbf{j})$ is modeled as,

$$I(\mathbf{j}) = \sum_i \rho_i \exp \left(-\frac{1}{2} (\mathbf{j} - \mathbf{p}_i)^T \Sigma_i^{-1} (\mathbf{j} - \mathbf{p}_i) \right), \quad (1)$$

where ρ_i denotes intensity, $\mathbf{p}_i$ the positions of centers, and Σ_i the covariances —all learnable parameters.

Methodology

- Step 1: Template Initialization.** Fit 500 Gaussian components to a group-average brain of 150 subjects to capture baseline geometry.
- Step 2: Structural Anchoring.** Lock the 500 Gaussian centers (μ_k) to act as fixed anatomical landmarks, guaranteeing exact spatial correspondence across all subjects.
- Step 3: Subject-Specific Adaptation.** Extract individual brain features by updating only the mixing weights (π_k) and covariances (Σ_k) using Weighted EM, initialized from the group-average baseline (to make parameter extraction deterministic).
- Step 4: Regression & Ensembling.** Subject-specific features (π_k, Σ_k) are fed directly into an ensemble of Lasso and XGBoost models.

Dataset

- VBM data from OpenBHB
- 3227 training subjects, 757 test subjects

The Engine: Weighted EM

- Voxels as Data:** Each voxel coordinate (x_j, y_j, z_j) acts as a spatial data point for the algorithm.
- Beyond Standard EM:** Traditional EM purely clusters spatial coordinates, ignoring the underlying tissue density.
- Adding Weights:** We utilize the MRI intensity $I(x_j, y_j, z_j)$ to serve as a physical weight for each voxel location.
- Modified M-Step:** Voxel responsibilities are multiplied by this weight during updates, ensuring the derived mixing weights (π_k) and covariances (Σ_k) capture true regional volume and shape.

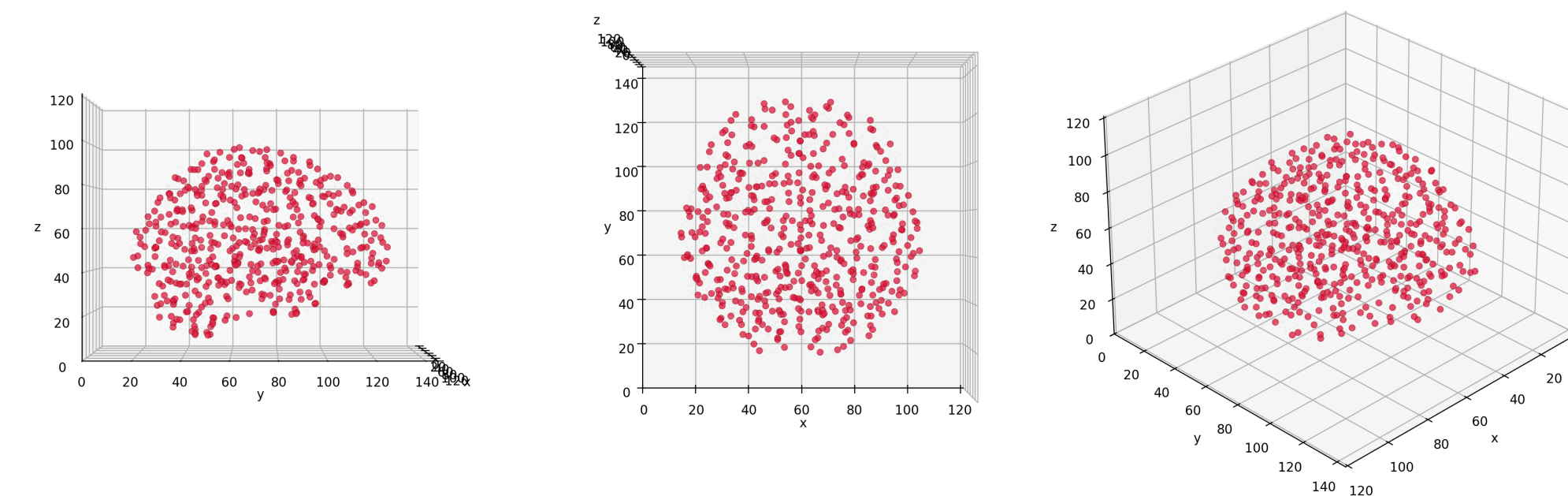


Figure 1. Spatial distribution of Gaussians

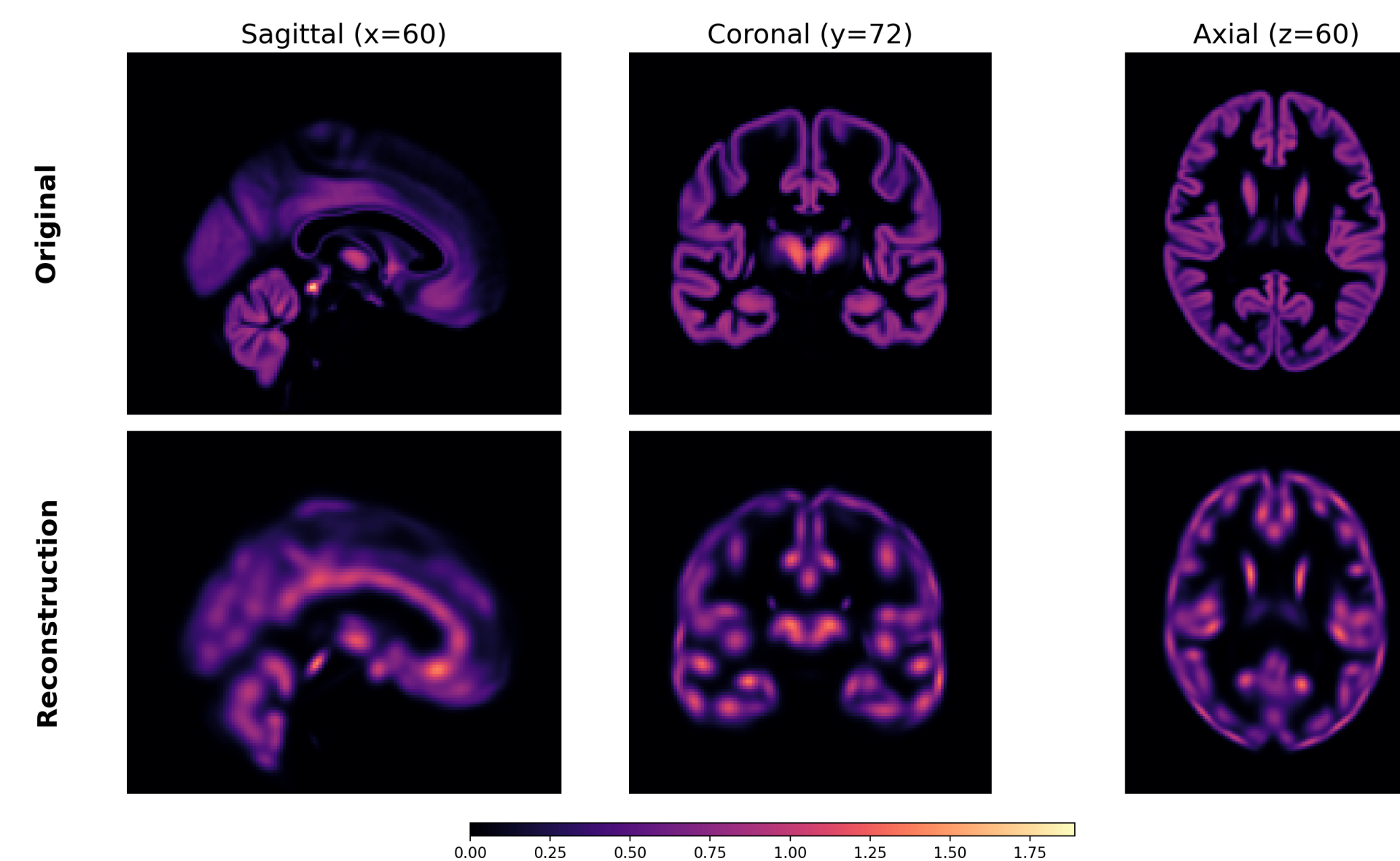


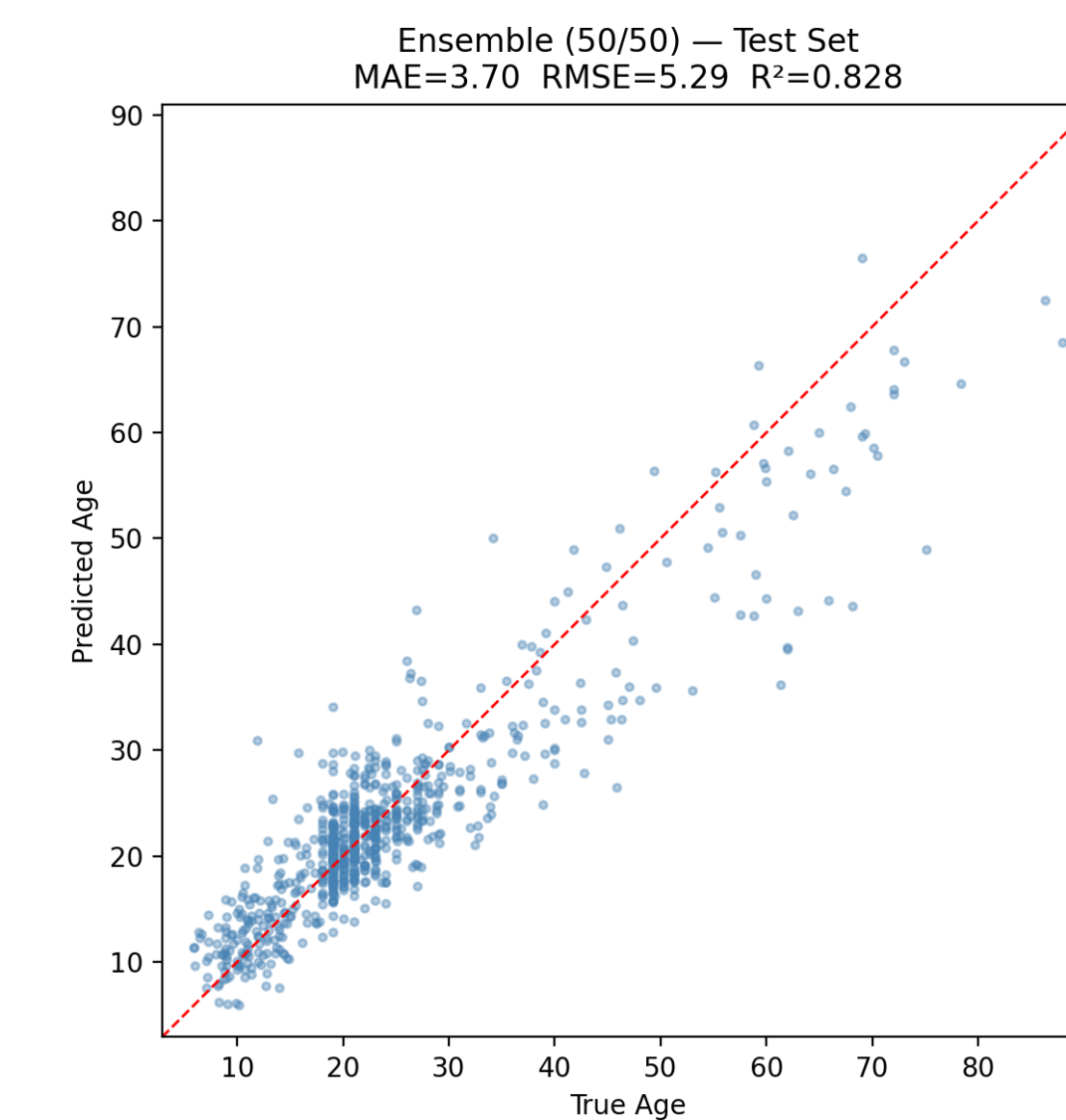
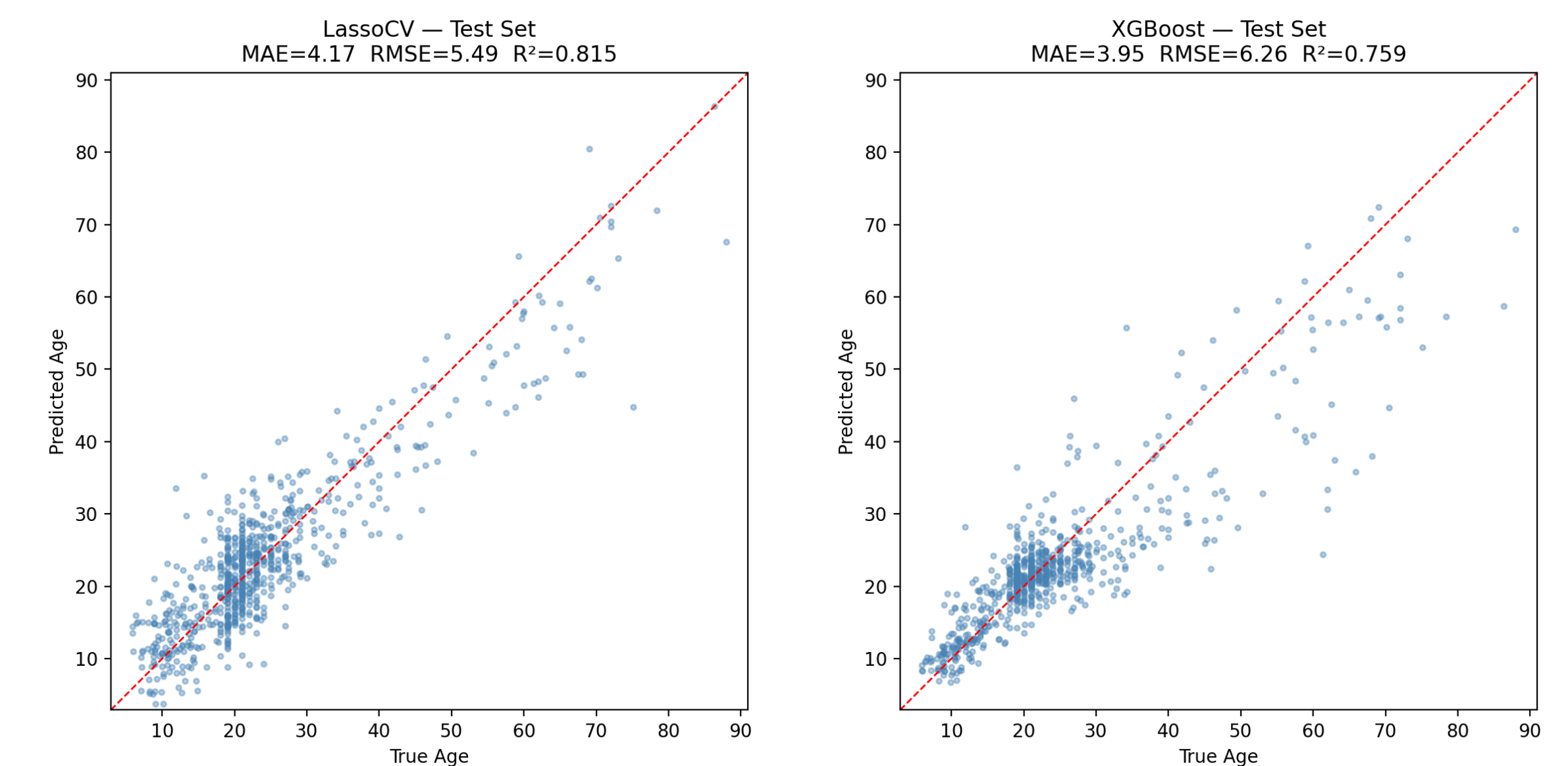
Figure 2. Original vs Reconstructed Average Brain

Regression

- L^1 Lasso Regression:** We need model sparsity as we have 3500 features, but the training data only contains 3227 subjects.
- XGBoost:** Generally performs better than linear regression by capturing some nonlinearities.
- Ensemble:** During validation, we found the errors of XGBoost and Linear regression were not highly correlated (correlation coefficient 0.5470), which justifies the use of ensembling.

Results

- In LassoCV, only 777 features survive.
- Predictions skew toward younger ages, likely due to age imbalance in the training data and/or higher biological variance in older brains.
- Despite operating in an underdetermined regime (3500 features, 3227 training samples), training and test MAE are close (3.35 years on the training data with 10-fold CV vs 3.7 years on test), indicating the model generalizes well without overfitting.



	Regression Model	MAE	FLOPs
Benchmark	ResNet(512)	2.67	~70 G
Our methods	Linear	4.17	263.2 G
	XGBoost	3.95	
	Ensemble	3.70	

References

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